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DBMS ASSIGNMENT - 2
2401010102 B.tech CSE SECA

## Sec-A

**Ques** Fill in the blanks.

1) The Entity Relationship Diagram usually represents new entities relate in a database offering a clear blueprint of the data's organizational structure.

2) In the Relational Model data is organized into tables with relationships b/w entities defined through foreign key constraints.

3) The object oriented model treat data as a collection of objects, each with unique attributes and methods, mirroring real world entities and interaction.

4) Integrity constraints ensure the reliability of data by enforcing rules that must be adhered to maintain the accuracy of information.

## Section - B

**Ques ②** In a health case system, how data abstraction can simplify the interaction b/w healthcare professionals and patient records.

**Ans** Data abstraction means hiding complex database detail and showing only important information to users. In a health care system, it helps doctor nurses and staff access patient records easily

There are three level of data abstraction:

① Physical level: Describes how data is stored in database. Users do not interact with this level

② Logical level: Defines what data is stored such as patient detail, doctor information and appointment and the relationship b/w them.

③ View Level: Show only required information to different users. For example, doctor can see medical history nurses can see patient details.

Ques③ Explore the potential benefits and challenges associated with implementing data abstraction in this context.

Ans Benefits:

① Improves data security & privacy.

② Makes the system easy to use for healthcare staff.

③ Provides data independence, so changes in database structure do not affect users.

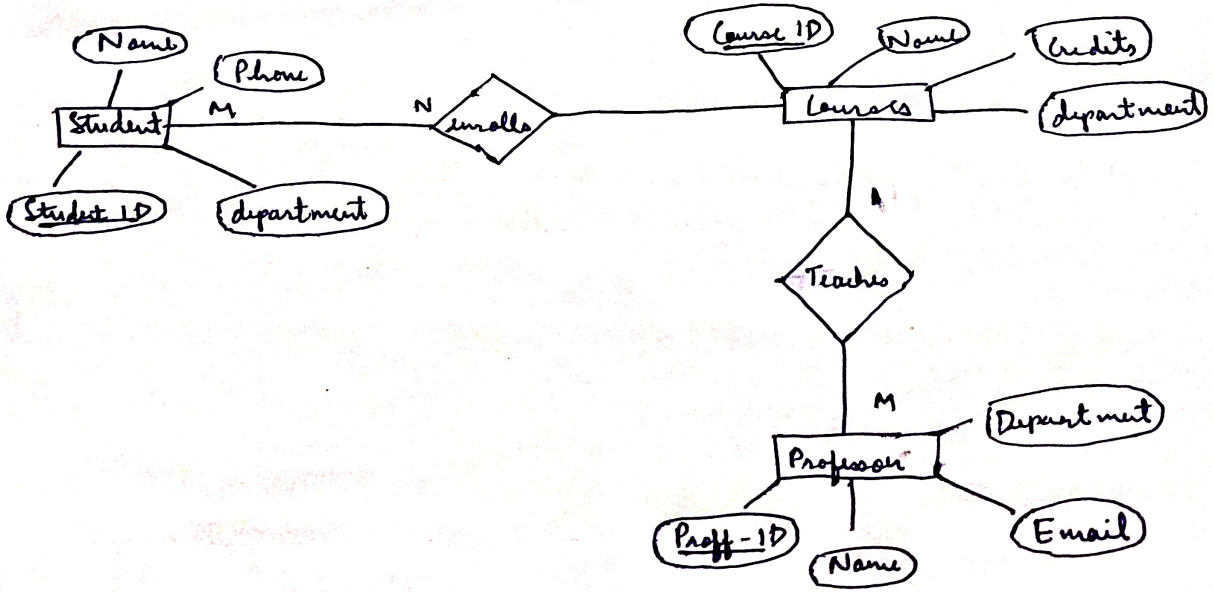
④ Helps in better data management. challenges

- Implementation complexity
- Performance overload
- High development cost.
- security risks if poorly implemented
- Training requirements

SECTION-C.

Ques (1) Designe a database for a university incorporating entities like students, courses, and professors using the entity relationship model, illustrate and explain the relationship b/w these entities - include attributes for each entity to capture relevant information such as student names, course, details & professor information.

Ans Student (student - ID PK), Name, Email, phone, Department)
Course (Course - ID PK), Name, credits, Department)
Attribute (Proff - ID PK), Name, Department)

Relationship.

① Student - Enroll - Course m : m
② Proff - Teaches - Course - 1 : m